

Stephanie J. Spielman, PhD

DATA SCIENTIST

Childhood Cancer Data Lab, Alex's Lemonade Stand Foundation 3 Bala Plaza, Bala Cynwyd, PA 19004

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I am a computational biologist and data scientist focused on developing open-source products to accelerate pediatric cancer research.

Education

The University of Texas at Austin	Austin, Texas, USA
PH.D., INTEGRATIVE BIOLOGY	2016
Brown University	Providence, RI, USA
B.S., BIOLOGY WITH HONORS	2010

Positions

Alex's Lemonade Stand Foundation	Bala Cynwyd, PA, USA
DATA SCIENTIST, CHILDHOOD CANCER DATA LAB	2022–Present
Developed open-source tools and data repositories, and lead training workshops to support pediatric cancer researchers	
Rowan University	Glassboro, New Jersey, USA
ASSISTANT PROFESSOR	2018–2022
Temple University	Philadelphia, Pennsylvania, PA
RESEARCH ASSISTANT PROFESSOR	2016–2018
Developed research in methods for quantifying protein evolution and taught graduate-level courses in Biostatistics and Evolutionary Medicine	
The University of Texas at Austin	Austin, Texas, USA
GRADUATE RESEARCH ASSISTANT	2011–2016
Computational molecular evolution research, including phylogenetic modeling and virus evolution.	
The University of Texas at Austin	Austin, Texas, USA
GRADUATE TEACHING ASSISTANT	2011–2016

- Supported undergraduate level courses in Biostatistics, Computational Biology and Bioinformatics, and Evolutionary Biology

Honors and Awards

President's Award for Excellence in Innotative Instructional Delivery	Glassboro, NJ, USA
ROWAN UNIVERSITY	2021
Awarded by the Faculty Center for Teaching Excellence	
Outstanding Dissertation Award	Austin, Texas, USA
THE UNIVERSITY OF TEXAS AT AUSTIN	2016
Awarded by the Office of Graduate Studies to the top dissertation among all of science, math, and engineering	
Ruth L. Kirschstein National Research Service Award (F31)	Austin, Texas, USA
THE UNIVERSITY OF TEXAS AT AUSTIN	2015
Pre-doctoral fellowship awarded by NIH/NIGMS	
Outstanding Teaching Award	Austin, Texas, USA
THE UNIVERSITY OF TEXAS AT AUSTIN	2014

- Awarded by the Biology Instructional Office

Selected Publications

- Hawkins, A. G., Shapiro, J. A., Spielman, S. J., Mejia, D. S., Prasad, D. V., Ichihara, N., Yakovets, A., Wheeler, K. G., Bethell, C. J., Foltz, S. M., O'Malley, J., Greene, C. S., & Taroni, J. N. (2024). The single-cell pediatric cancer atlas: Data portal and open-source tools for single-cell transcriptomics of pediatric tumors. *bioRxiv*. <https://doi.org/10.1101/2024.04.19.590243>

2. Shapiro, J. A., Gaonkar, K. S., Spielman, S. J., Savonen, C. L., Bethell, C. J., Jin, R., Rathi, K. S., Zhu, Y., Egolf, L. E., Farrow, B. K., Miller, D. P., Yang, Y., Koganti, T., Noureen, N., Koptyra, M. P., Duong, N., Santi, M., Kim, J., Robins, S., ... Taroni, J. N. (2023). OpenPBTA: The open pediatric brain tumor atlas. *Cell Genomics*, 3(7), 100340. <https://doi.org/10.1016/j.xgen.2023.100340>
3. Spielman, S. J., & Miraglia, M. L. (2021). Relative model selection of evolutionary substitution models can be sensitive to multiple sequence alignment uncertainty. *BMC Ecology and Evolution*, 21, 214. <https://doi.org/10.1186/s12862-021-01931-5>
4. Spielman, S. J. (2020). Relative model fit does not predict topological accuracy in single-gene protein phylogenetics. *Molecular Biology and Evolution*, 37(7), 2110–2123. <https://doi.org/10.1093/molbev/msaa075>
5. Spielman, S. J., & Moore, E. K. (2020). dragon: A new tool for exploring redox evolution preserved in the mineral record. *Frontiers in Earth Science*. <https://doi.org/10.3389/feart.2020.585087>
6. Spielman, S. J., & Kosakovsky Pond, S. L. (2018). Relative evolutionary rates in proteins are largely insensitive to the substitution model. *Molecular Biology and Evolution*, 35(9), 2307–2317. <https://doi.org/10.1093/molbev/msy127>
7. Echave, J., Spielman, S. J., & Wilke, C. O. (2016). Causes of evolutionary rate variation among protein sites. *Nature Reviews Genetics*, 17(2), 109–121. <https://doi.org/10.1038/nrg.2015.18>
8. Meyer, A. G., Spielman, S. J., Bedford, T., & Wilke, C. O. (2015). Time dependence of evolutionary metrics during the 2009 pandemic influenza virus outbreak. *Virus Evolution*, 1(1), vev006. <https://doi.org/10.1093/ve/vev006>
9. Spielman, S. J., & Wilke, C. O. (2015). pyvolve: A flexible python module for simulating sequences along phylogenies. *PLOS ONE*, 10(9), e0139047. <https://doi.org/10.1371/journal.pone.0139047>
10. Spielman, S. J., & Wilke, C. O. (2015). The relationship between dN/dS and scaled selection coefficients. *Molecular Biology and Evolution*, 32(4), 1097–1108. <https://doi.org/10.1093/molbev/msv003>